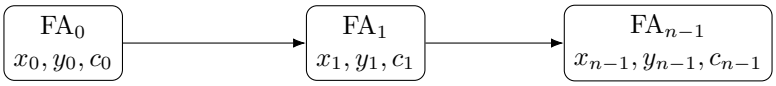
VLSI Testing: 2013 Midterm

Reconstructed questions and independently derived solutions. Personal names, identifiers, handwritten answers and administrative headers are omitted. Figures are redrawn as logical diagrams or explicit netlists. A source-reading note identifies a meaningful ambiguity instead of silently changing the question. Finite Boolean checks are supplied in the examination collection's Python verification script.

1 Single-cell faults in a ripple-carry adder

Question. An n-bit ripple-carry adder is built from full-adder cells. Derive a test set covering every single cell fault with a size independent of n. The cell outputs are sum and carry.



Solution. A single faulty cell may implement any incorrect truth table, so apply all eight (x_i, y_i, c_i) triples to every cell. Use the following eight whole-adder tests:

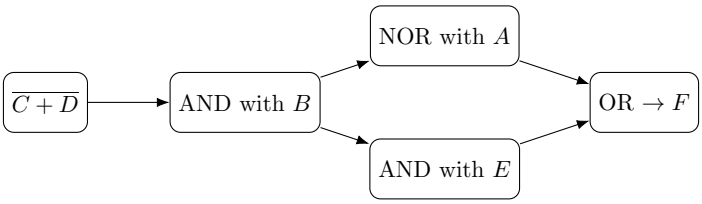
1. Four tests: every bit has $(x_i, y_i) = 01$, with $c_0 = 0$ and 1; then every bit has $(x_i, y_i) = 10$, again with $c_0 = 0$ and 1. Both pairs propagate carry, so every cell sees the selected carry.
2. One test: all pairs are 00 and $c_0 = 0$. One test: all pairs are 11 and $c_0 = 1$.
3. One test alternates $00, 11, 00, 11, \ldots$ from the least significant cell, with $c_0 = 1$. The other alternates $11, 00, 11, 00, \ldots$, with $c_0 = 0$. Thus every cell sees both 001 and 110 across these two tests.

The eight local triples are now covered at every position. A wrong sum is directly observable. A wrong carry changes the next good cell's sum (sum is XOR in carry); at the last cell it is the primary carry output. Therefore one faulty cell cannot hide its error under this model. The number of tests is eight, independent of n. This argument assumes all sum bits and the final carry are observed and only one cell is faulty.

2 ATPG by the D-algorithm and single-path sensitization

Question. Generate a test for $x/0$ using the D-algorithm and a test for $y/0$ using single-path sensitization, for the following circuit.

Signal	Defining expression
u	$\overline{C + D}$
y	Bu
h	$\overline{A + y}$
i	yE
F	$h + i$



Solution. The marked x is the NOR output u and y is the AND output. For $x/0$, excite $u = 1$ with $C = D = 0$, set $B = 1$, and obtain $y = D$, where $D = (1, 0)$ denotes good/faulty values. Set $A = 0$ and $E = 0$: $h = \bar{D}$, $i = 0$, $F = \bar{D}$. A complete vector is $(A, B, C, D, E) = 01000$. For $y/0$, the same vector activates $y = 1$, blocks the lower reconvergent path with $E = 0$ and sensitizes the path $y \rightarrow h \rightarrow F$ with $A = 0$. It produces $(F_{\text{good}}, F_{\text{faulty}}) = (0, 1)$ in both cases.

3 Combinational SCOAP

Question. Compute controllability and observability for every signal in the following netlist.

Use $CC_0 = CC_1 = 1$ at a primary input and $CO = 0$ at a primary output; each gate adds one. For an AND gate, $CC_0 = 1 + \min CC_0(i)$ and $CC_1 = 1 + \sum CC_1(i)$. For an OR gate, interchange the roles of 0 and 1. NAND/NOR swap the two output controllabilities; an inverter swaps its input values and adds one. To observe one gate input, set the other inputs to their noncontrolling values, add their controllability costs, add CO of the output and add one. At a fanout stem take the minimum branch CO. For XOR/XNOR, minimize the sum of input costs subject to the desired parity. These are structural SCOAP estimates; reconvergent inputs are treated independently.

Signal	Defining expression
N	$\overline{X_2 X_3}$
E	$X_1 N$
M	$\overline{N X_4}$
I	$\bar{X}_3$
L	$I X_4$
F	$E + M + L$

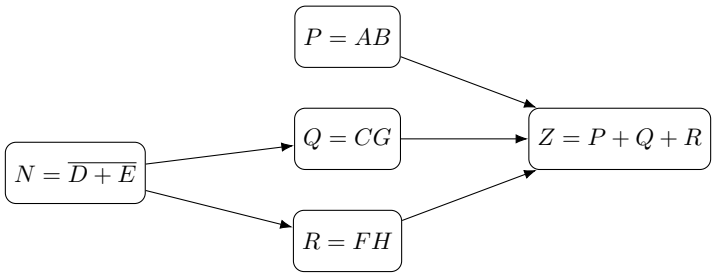
Both E and F are primary outputs. In particular, $CO(E) = 0$ and N is observed through E more cheaply than through F.

Signal	CC_0	CC_1	CO
$X1$	1	1	3
$X2$	1	1	4
$X3$	1	1	4
$X4$	1	1	8
N	3	2	2
E	2	4	0
M	4	2	5
I	2	2	9
L	2	4	7
F	9	3	0

For example, $CC_0(F) = 2 + 4 + 2 + 1 = 9$ and $CO(N) = CO(E) + CC_1(X_1) + 1 = 2$. Branch observabilities are $N \rightarrow E : 2$, $N \rightarrow M : 7$, $E \rightarrow F : 7$, $X_3 \rightarrow N : 4$, $X_3 \rightarrow I : 10$, $X_4 \rightarrow M : 8$, $X_4 \rightarrow L : 10$.

4 Deductive fault simulation

Question. The circuit is $N = \overline{D+E}$, $P = AB$, $Q = CG$, $R = FH$, $Z = P + Q + R$, where G and H are separate branches of N. The four faults are $f_A = A/1$, $f_E = E/0$, $f_G = G/1$, $f_H = H/1$. Find detected faults for $(A, B, C, D, E, F) = 011011$ and 011111.



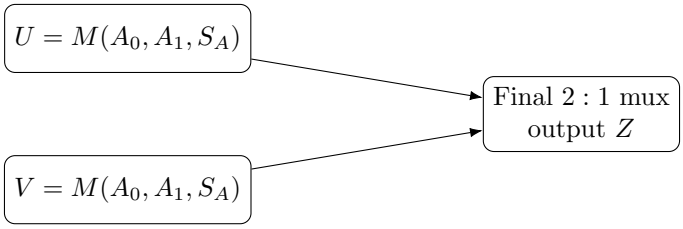
Solution. Both patterns have $(P, Q, R) = (0, 0, 0)$ and $Z = 0$. For 011011, $D = 0, E = 1$, so $E/0$ changes N from 0 to 1. Deductive lists are

$$L(P) = \{f_A\}, \quad L(N) = \{f_E\}, \quad L(Q) = \{f_E, f_G\}, \quad L(R) = \{f_E, f_H\}.$$

All OR inputs are zero, so take the union: $L(Z) = \{f_A, f_E, f_G, f_H\}$. For 011111, $D = 1$ masks $E/0$ at the NOR gate. Consequently $L(N) = \varnothing$, $L(Q) = \{f_G\}$, $L(R) = \{f_H\}$ and $L(Z) = \{f_A, f_G, f_H\}$. The branches G, H must be kept distinct; a stem fault is a different fault.

5 Fault tests for a mux network

Question. Find two tests for the marked SA0 and SA1 faults.



Solution. Let $U = M(A_0, A_1, S_A)$ and $V = M(A_0, A_1, S_A)$, where $M(a, b, s) = \bar{s}a + sb$. The final mux is selected by $\bar{S}$. The marked faults are $U/0$ and $\bar{S}/1$. For $U/0$, choose $\bar{S} = 0$, $S_A = 0, A_0 = 1$; for example $(A_0, A_1, S_A, \bar{S}) = 1000$ gives $Z = 1/0$.

Source-reading consequence. The final-select fault $\bar{S}/1$ is unobservable because $U = V$ for every input. No second test exists for that exact circuit. If the fault was intended on a select branch inside one of the first two muxes, that is a different fault, as in the 2014 version.

6 Short questions on manufacturing test

(a) **Verification and test.** Verification asks whether the design implements the specification. Manufacturing test applies stimuli to fabricated hardware and checks for physical defects represented by a fault model.

(b) **Yield and defect level.** With yield $Y = 0.5$, use the model $DL = 1 - Y^{1-FC}$ for the defect fraction among shipped parts. For $DL \leq 100 \times 10^{-6}$,

$$FC \geq 1 - \frac{\ln(0.9999)}{\ln(0.5)} = 0.99985572\ldots,$$

so at least 99.9856% fault coverage is required *under this model*. Yield alone cannot determine defect level without a coverage/escape model.

(c) **Test flow.** Select a defect/fault model; generate and simulate tests; prepare tester programs and expected responses; apply tests at wafer/package/board stages as appropriate; compare responses, diagnose failures and bin devices; feed results back to design and manufacturing.

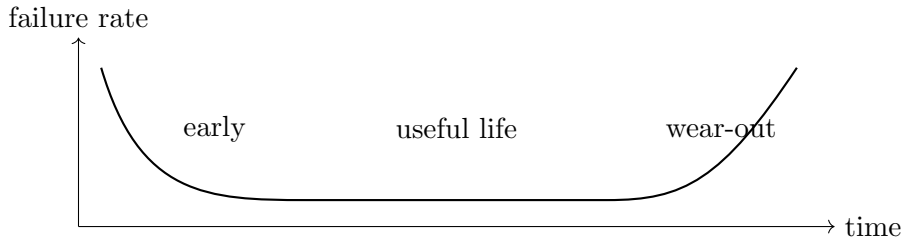
(d) **Detection, location, diagnosis.** Detection distinguishes faulty from fault-free behavior. Location narrows the physical or logical site. Diagnosis infers likely fault mechanisms/sites from the response syndrome.

(e) **Fault modeling.** A finite logical/electrical abstraction of defects makes test generation and coverage measurable; model coverage need not equal defect coverage.

(f) **Bathtub curve.** Early failures decrease as weak parts fail; useful life has a low approximately constant hazard; wear-out failures increase with aging.

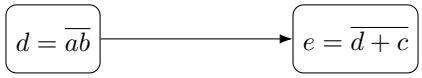
(g) **Fault coverage.** $FC = N_{\text{detected}}/N_{\text{modeled}}$, with the denominator and any exclusion of untestable faults stated explicitly.

(h) **Fault simulation.** Predict faulty responses, measure coverage, identify undetected faults and grade/compact tests; optionally provide diagnostic signatures.



7 Equivalence and dominance

Question. For $d = \overline{ab}$ and $e = \overline{d + c}$, classify $a/0$, $b/1$, $d/1$ and $e/0$ by equivalence and dominance.



Solution. Since $e = ab\bar{c}$, the detection sets, in (a, b, c) order, are

$$T(a/0) = T(d/1) = T(e/0) = \{110\}, \qquad T(b/1) = \{100\}.$$

Thus the first three faults are equivalent. The fourth is neither equivalent to nor in a strict dominance relationship with them: the two nonempty detection sets are disjoint. Stating the sets also avoids opposite conventions for dominance.